Phase 1 - Arrondissement Data Exploration

Notebook d'exploration de la table `master_arrondissements` dans la voie pedagogique officielle du projet.

1. Setup & Imports

In [1]:
```
from pathlib import Path
import sys

import geopandas as gpd
import matplotlib.pyplot as plt
import pandas as pd

project_root = Path.cwd().resolve()
if project_root.name == "notebooks":
    project_root = project_root.parent

if str(project_root) not in sys.path:
    sys.path.insert(0, str(project_root))

from src.modeling import build_pedagogical_master_table

plt.style.use("seaborn-v0_8-whitegrid")
```

2. Data Loading

In [2]:
```
master_arr = build_pedagogical_master_table()
master_arr.head()
```

Out[2]:

	arrondissement_code	arrondissement_name	geometry	y_bin_count	x1_population	x2_
0	01	Paris 1er Arrondissement	POLYGON ((2.32145 48.86372, 2.32147 48.86375, ...	333	15475.0	
1	02	Paris 2e Arrondissement	POLYGON ((2.34256 48.8654, 2.34217 48.8655, 2....	75	20433.0	
2	03	Paris 3e Arrondissement	POLYGON ((2.35674 48.86015, 2.35631 48.86029, ...	114	32772.0	
3	04	Paris 4e Arrondissement	POLYGON ((2.34707 48.85718, 2.34711 48.85724, ...	161	28039.0	
4	05	Paris 5e Arrondissement	POLYGON ((2.35145 48.83684, 2.35122 48.83686, ...	318	55925.0	

3. Data Quality Check

Verification des types, des valeurs manquantes et de la cardinalite des 20 arrondissements.

In [3]:

```
pd.DataFrame({
    "dtype": master_arr.drop(columns="geometry").dtypes.astype(str),
    "na_count": master_arr.drop(columns="geometry").isna().sum(),
})
```

Out[3]:

	dtype	na_count
arrondissement_code	object	0
arrondissement_name	object	0
y_bin_count	int64	0
x1_population	float64	0
x2_commerce_restaurant_count	int64	0
x3_transport_station_count	int64	0
x4_green_area_m2	float64	0
x5_road_length_km	float64	0

In [4]:
```
master_arr[["arrondissement_code", "arrondissement_name"]].sort_values("arrondissem
```

Out[4]:

	arrondissement_code	arrondissement_name
0	01	Paris 1er Arrondissement
1	02	Paris 2e Arrondissement
2	03	Paris 3e Arrondissement
3	04	Paris 4e Arrondissement
4	05	Paris 5e Arrondissement
5	06	Paris 6e Arrondissement
6	07	Paris 7e Arrondissement
7	08	Paris 8e Arrondissement
8	09	Paris 9e Arrondissement
9	10	Paris 10e Arrondissement
10	11	Paris 11e Arrondissement
11	12	Paris 12e Arrondissement
12	13	Paris 13e Arrondissement
13	14	Paris 14e Arrondissement
14	15	Paris 15e Arrondissement
15	16	Paris 16e Arrondissement
16	17	Paris 17e Arrondissement
17	18	Paris 18e Arrondissement
18	19	Paris 19e Arrondissement
19	20	Paris 20e Arrondissement

4. Descriptive Statistics

Les variables `X1..X5` et la cible `Y` sont analysees a l'echelle arrondissement.

In [5]:

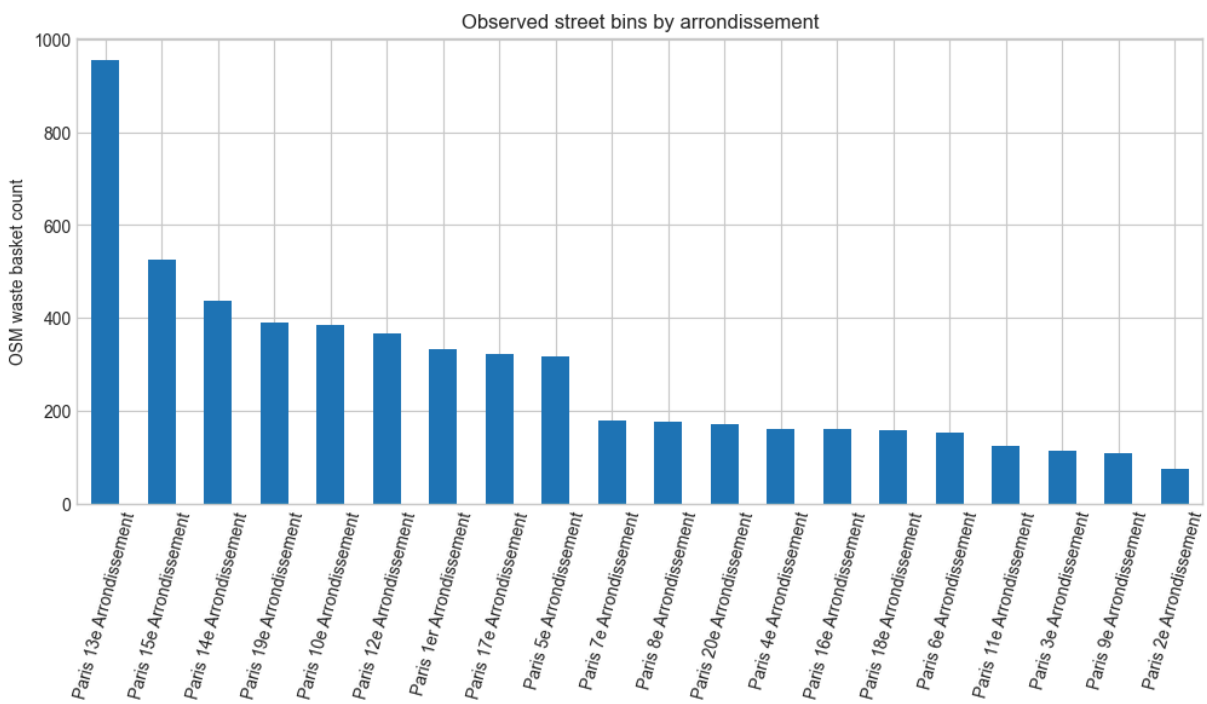
```
analysis_columns = [
    "y_bin_count",
    "x1_population",
    "x2_commerce_restaurant_count",
    "x3_transport_station_count",
    "x4_green_area_m2",
    "x5_road_length_km",
]
master_arr[analysis_columns].describe().T
```

Out[5]:

	count	mean	std	min	2
y_bin_count	20.0	2.807500e+02	2.042619e+02	75.000000	157.000
x1_population	20.0	1.056852e+05	6.983243e+04	15475.000000	39178.500
x2_commerce_restaurant_count	20.0	2.242550e+03	5.777329e+02	1259.000000	1888.500
x3_transport_station_count	20.0	2.440000e+01	1.397893e+01	5.000000	12.750
x4_green_area_m2	20.0	1.236111e+06	2.940033e+06	4517.143029	50387.394
x5_road_length_km	20.0	1.160652e+02	7.108781e+01	28.105616	61.212

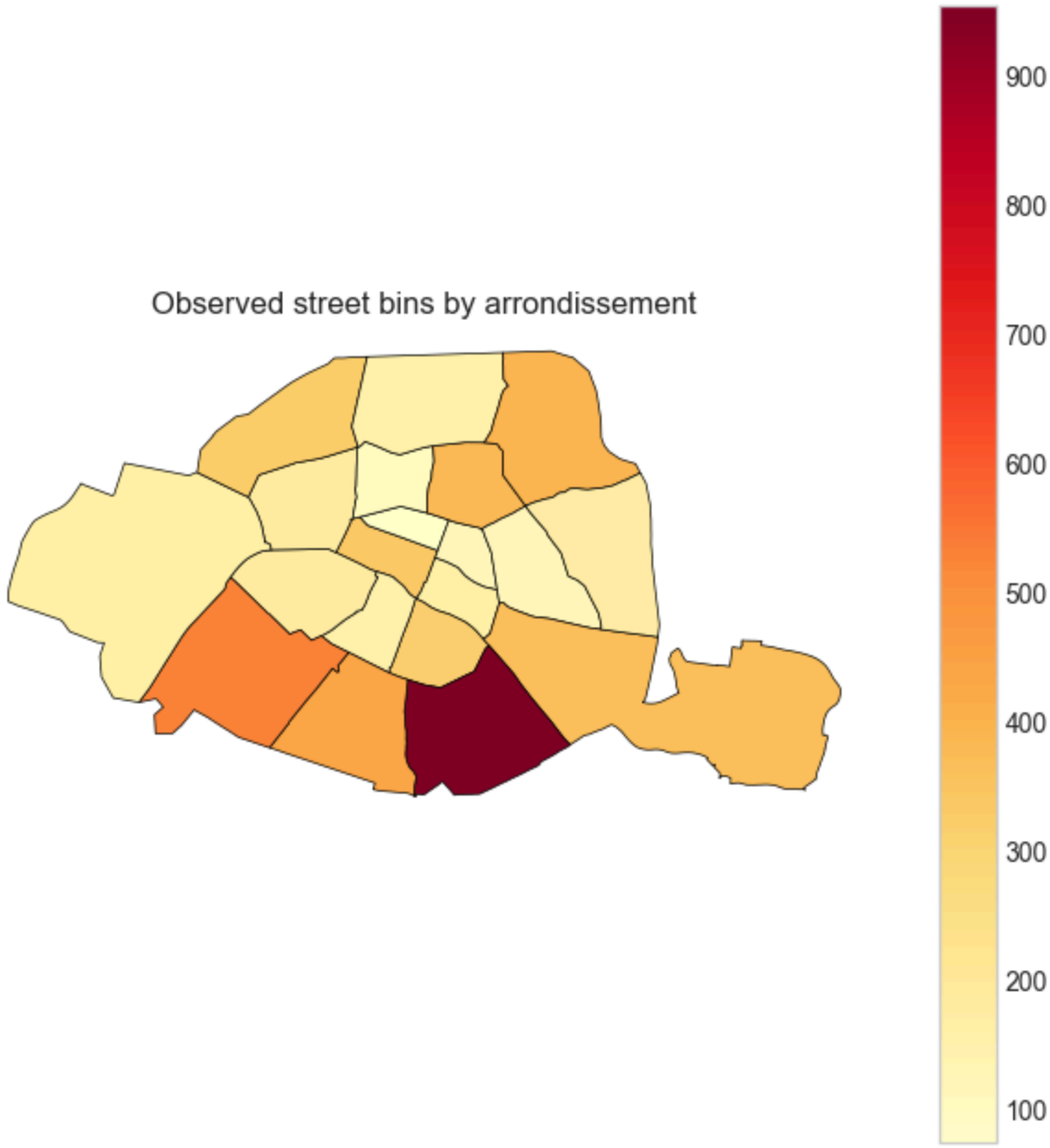
In [6]:

```
plot_df = master_arr.sort_values("y_bin_count", ascending=False)
ax = plot_df.plot(
    x="arrondissement_name",
    y="y_bin_count",
    kind="bar",
    figsize=(12, 5),
    legend=False,
    color="#1f77b4",
)
ax.set_title("Observed street bins by arrondissement")
ax.set_xlabel("")
ax.set_ylabel("OSM waste basket count")
plt.xticks(rotation=75)
plt.show()
```



5. Geographic Visualization

In [7]:
```
ax = master_arr.plot(
    column="y_bin_count",
    cmap="YlOrRd",
    figsize=(8, 8),
    legend=True,
    edgecolor="black",
    linewidth=0.5,
)
ax.set_title("Observed street bins by arrondissement")
ax.set_axis_off()
plt.show()
```



6. Correlation Analysis

In [8]:
```
master_arr[analysis_columns].corr(numeric_only=True)
```

Out[8]:

	y_bin_count	x1_population	x2_commerce_restaurant_count
y_bin_count	1.000000	0.464502	0.078704
x1_population	0.464502	1.000000	0.383387
x2_commerce_restaurant_count	0.078704	0.383387	1.000000
x3_transport_station_count	0.570339	0.876210	0.300006
x4_green_area_m2	0.014852	0.286902	-0.083104
x5_road_length_km	0.393866	0.781218	0.060674

7. Key Findings & Next Steps

- La table primaire contient 20 observations, une par arrondissement.
- La cible `y_bin_count` et les variables `X1..X5` sont deja alignees avec la consigne pedagogique.
- L'etape suivante consiste a appliquer `KMeans`, creer `cl_2` et `cl_3`, puis ajuster la regression lineaire multiple.